Lesson 4: How Stiff Is a Spring?

Unit

Energy: Stores and Transfers

Lesson focus

Students stretch a spring, plot force against extension and find its spring constant k, using F = k e. The point of k is Lesson 5: it is the number that tells us how much energy a stretched spring stores.

Specification

AQA GCSE Physics 4.5.3 Forces and elasticity (taught here ahead of the Forces unit). The specification asks students to recall and apply F = k e; it is also on the AQA Physics equation sheet.

Scope

Students calculate F, and rearrange to find k (k = F ÷ e) or e (e = F ÷ k). They plot two graphs of the same results: extension against force first, then force against extension, whose gradient is k.

Learning objectives

By the end of the lesson, students should be able to:

1. Explain why two forces are needed to stretch a spring.
2. Describe the difference between elastic and inelastic deformation.
3. Calculate extension from the stretched and original lengths.
4. Use F = k e to calculate a force or a spring constant.
5. Plot extension against force, then force against extension; explain why they look different; find k from the gradient of the second; identify the limit of proportionality.

Key vocabulary

Keyword	Student-friendly definition
Extension (e)	How much longer the spring is than its original length, in metres
Spring constant (k)	The force needed to stretch the spring by 1 m, in newtons per metre (N/m); a stiffer spring has a bigger k
Elastic deformation	The object returns to its original shape when the force is removed
Inelastic deformation	The object stays stretched or bent when the force is removed
Directly proportional	Doubling one quantity doubles the other; the graph is a straight line through the origin
Limit of proportionality	The point beyond which force and extension are no longer proportional

The equation

Quantity	Symbol	Unit
Force	F	newton (N)
Spring constant	k	newton per metre (N/m)
Extension	e	metre (m)

$$F = k\,e$$
force = spring constant × extension

Rearranged: k = F ÷ e, and e = F ÷ k.

Hooke's law: the extension of a spring is directly proportional to the force on it, provided the limit of proportionality is not exceeded.

Turning masses into forces at the bench

Recall from Lesson 3: gravity pulls each kilogram down with 9.8 N. So each 100 g pulls down with 0.98 N.

> *At the bench, we round this: each 100 g mass pulls down with about 1 N.*

Use this rule **only** at the bench. Students record force in newtons, never in grams. A mass is not a force; if students plot grams against centimetres, their k has the wrong unit and Lesson 5 will fail.

Equipment and resources (per group)

- Clamp stand, boss and clamp, with a heavy weight or G-clamp on the base
- Spring, mass hanger and 100 g slotted masses (up to 700 g in total)
- Metre rule clamped vertically beside the spring
- A pointer (a pin or a strip of card taped to the bottom of the spring)
- Safety goggles

Safety notes

- Wear goggles: a spring that slips off the hook can whip.
- Stand while you work; keep feet out from under the masses.
- Clamp the stand so it cannot tip.
- Add masses gently; do not let them drop onto the hanger.

Starter: One force or two?

Hang a spring from a clamp and pull the bottom.

Ask: what would happen if only my hand pulled and the clamp did not hold the top? (The spring would move, not stretch.) To change an object's shape while it stays still, at least two forces must act on it: here, my pull down and the clamp's pull up.

Then stretch a spring gently and let go (it returns: elastic). Over-stretch an old spring until it stays longer (inelastic). Keep the damaged spring to show later.

Main teaching: Extension, not length

> *extension = stretched length − original length*

Draw the spring before and after, with the original length, stretched length and extension marked. Students most often record the length instead of the extension; the graph then does not pass through the origin.

Practical: Force and extension

1. Clamp the spring and metre rule; line the pointer up with the scale. Record the original length (with the empty hanger, counted as the zero load, or with the hanger's mass included: decide as a class).

2. Add 100 g. Record the new length.
3. Repeat in 100 g steps up to 700 g.
4. For each load, calculate the force (about 1 N per 100 g) and the extension.
5. Remove the masses one at a time and check that the spring returns to its original length.

Sample results (original length 5.0 cm):

Mass added / g	Force / N	Length / cm	Extension / cm
0	0	5.0	0.0
100	1	9.0	4.0
200	2	13.1	8.1
300	3	17.0	12.0
400	4	21.0	16.0
500	5	25.1	20.1
600	6	29.0	24.0
700	7	34.5	29.5

Two graphs of the same results

Graph 1: extension against force. Force is the independent variable (we choose it by adding masses), so, as in Unit 1, it goes on the x-axis and extension on the y-axis. Students plot the points and draw a straight best-fit line through the origin for the first six points.

> *This graph answers the question we asked: what happens to the extension when the force increases? It increases in proportion, up to the limit of proportionality.*

Graph 2: force against extension. Now swap the axes: extension on the x-axis, force on the y-axis, extension in metres.

> *k = gradient = change in force ÷ change in extension*
> *k = 6 N ÷ 0.240 m*
> *k = 25 N/m*

Explain the difference. Both graphs show the same results, so both show the same straight line through the origin, just reflected. What changes is the gradient:

Graph	y-axis	x-axis	Gradient means
1	extension	force	extension per newton (0.04 m per N): how stretchy the spring is
2	force	extension	force per metre, which is k: how stiff the spring is

Physicists draw graph 2 because its gradient is the spring constant, the number we need for elastic potential energy in Lesson 5. AQA exam questions usually show graph 2. A stiffer spring gives a steeper line on graph 2, but a shallower line on graph 1.

On both graphs the 700 g point lies off the line: the spring has passed its limit of proportionality. Check whether it returned to 5.0 cm; if not, it was also inelastically deformed.

Worked example (no graph):

> *A spring stretches by 4.0 cm when a force of 2.0 N acts on it. Calculate its spring constant.*
> *e = 4.0 cm = 0.040 m*
> *F = k e, so k = F ÷ e*
> *k = 2.0 ÷ 0.040*
> *k = 50 N/m*

Check for understanding

1. A spring is 6.0 cm long. With a load it is 9.5 cm long. What is the extension? (3.5 cm)
2. What does a spring constant of 50 N/m mean? (50 N would stretch it by 1 m, if it did not break first.)
3. Spring A has k = 20 N/m; spring B has k = 80 N/m. Which is stiffer?
4. Calculate the force needed to stretch a 40 N/m spring by 0.15 m. (6.0 N)
5. How can you tell from the graph where the limit of proportionality is?
6. A spring has k = 40 N/m. What extension does a 2.0 N force give? (0.050 m = 5.0 cm)
7. Spring A is stiffer than spring B. Whose line is steeper on a force–extension graph?

Plenary

Hold up two springs with different k. Ask: if I stretch both by 10 cm, which one did I push harder on? Which one do you think now stores more energy? Hold the question for Lesson 5.

Possible prep

Students finish both graphs and the spring constant if needed. Stretch: explain why a graph of length against force is a straight line but does not go through the origin.

Teacher notes

1. Keep each group's k. Lesson 5 uses it.
2. The bench rule makes k about 2% too high (each 100 g really pulls with 0.98 N). Mention it only if a student asks; it is a good example of a systematic error from Unit 1.
3. Centimetres to metres is the commonest error in calculating k. Insist on extension in metres before dividing.
4. "Linear" and "non-linear" are worth naming: the straight part of the graph is linear; beyond the limit of proportionality it is non-linear.